

Le Khanh Trinh Mai

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EDUCATION

Kahlert School of Computing, University of Utah

Salt Lake City, UT

B.S. in Computer Science; GPA: 3.8/4.0

Expected May 2027

- **Relevant Coursework:** Algorithms, Software Practice I & II, Mobile Application Programming, Intro to Practical ML, Data Analysis, Web Dev I, Database System, Coding with AI Agents

EXPERIENCE

Select Portfolio Servicing, Inc.

Sep 2025 – Dec 2025

Software Engineer Intern

Salt Lake City, UT

- Designed and built a Python-based static analysis platform for legacy C# applications, extracting 7+ code artifact types including classes, methods, API endpoints, ORM models, database relations, call sites, and service dependencies.
- Automated architecture discovery across 10+ internal engineering projects by converting raw source-code signals into normalized dependency graphs for cross-service, database, and business-logic analysis.
- Developed an interactive React Flow architecture explorer to visualize 3+ dependency layers, helping engineers reduce manual code tracing from hours of source-code inspection to minutes of graph-based navigation.
- Integrated Python microservices with a local LLM pipeline to generate consistent business-level documentation from extracted code context, improving onboarding and reducing reliance on manually maintained architecture notes.

Scientific Computing and Imaging Institute (University of Utah)

Aug 2024 – Dec 2025

Assistant Research Engineer

Salt Lake City, UT

- Built interactive, web-based analytics dashboards for computer-architecture experiments using D3.js and SvelteKit, with Python (Matplotlib/Seaborn) for exploratory analysis and figure prototyping.
- Partnered with simulation engineers (College of William & Mary) to transform high-volume raw performance telemetry (thousands of metrics across runs) into cleaned, structured datasets and actionable visualizations, reducing analysis turnaround by 80%.
- Improved visualization rendering throughput by 20% via data preprocessing, incremental updates, and DOM/Canvas optimization, enabling real-time interaction on large datasets.

InCaria - A Lifestyle Analysis Platform/Tech Startup

Jan 2025 – May 2025

Fullstack Software Engineer Intern

Salt Lake City, UT

- Developed a cross-platform mobile app using React Native with TypeScript, contributing 20% toward MVP completion by building core interface components and user workflows.
- Integrated the React Native front end with a secure Django/PostgreSQL backend via RESTful APIs, ensuring real-time synchronization in health and lifestyle data handling.

PROJECTS

Desktop Code Editor | Python, PySide6, Qt, pytest, pytest-qt

- Built a 6.1K+ line native desktop code editor with multi-tab editing, split panes, drag-and-drop tab movement, integrated file explorer workflows, and save-state aware document management.
- Improved large-file opening by 99.9% and replace-all performance by 99.6% on profiled workloads by introducing QThread-based async file I/O, 16 KB chunked document insertion, and incremental syntax rehighlighting.
- Implemented support for 25 file extensions across 7 language families and 6 persisted themes, adding syntax highlighting, regex/case-sensitive search, live match navigation, auto-indent, bracket matching, and quote/bracket auto-close behavior.
- Validated reliability with 709 automated tests across 18 pytest/pytest-qt suites and 96% coverage, hardening the editor against 100,000-line files, 6.2 MB search workloads, binary-file rejection, and cross-split drag-and-drop edge cases.

LMS: Role-Based Learning Management System | C#, ASP.NET Core MVC, Entity Framework Core, MySQL, ASP.NET Identity

- Built 24+ backend controller actions to support end-to-end LMS operations, including registration, department/course/class management, enrollment, assignment submission, grading, GPA calculation, and course catalog retrieval.
- Designed and integrated 2 MySQL-backed EF Core DbContexts to separate authentication data from academic records, improving data organization and supporting secure role-based access for 3 user types.
- Engineered business logic across 10 relational entities to manage departments, courses, classes, enrollments, assignments, submissions, and users through LINQ-driven CRUD and query workflows.
- Automated 100% of class letter-grade recalculations after assignment creation and submission grading, implementing weighted category logic, missing-submission handling, and persistent grade updates in the database.

sParky: Wildfire Prediction Web App | React, Node.js, AWS, OpenWeather

- Engineered a full-stack AI-powered web app with a RESTful Node.js backend predicting wildfire and air quality threats using live sensor data.
- Trained and deployed an ML model on AWS SageMaker with 78.6 % accuracy, estimating wildfire probability and spread radius across Utah's microclimates.
- Generated personalized health advisories and hyper-local forecasts for 50+ locations daily via AWS Bedrock, powered by real-time OpenWeather AQI inputs.
- Enabled dynamic wildfire risk visualization using AWS Location Services, increasing platform engagement by 60% through interactive, map-based insights.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C#, C++, Kotlin

Web Development: PERN/MERN stack, React, SvelteKit, Blazor

Databases: SQL, PostgreSQL, MongoDB, MySQL, SQLite, Supabase

Tools & Other: Git, Docker, REST APIs